

AI Governance Interoperability Index

Mapping Fragmentation and Pathways for Inclusive Global AI Governance

Policy Brief

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Executive Summary

Artificial intelligence is increasingly shaping public services, economies, education, health systems, labor markets, security, information ecosystems, and international development. As AI systems expand across borders, governments and institutions are developing different approaches to govern their risks and opportunities. Some approaches rely on binding regulation, while others use voluntary principles, standards, advisory frameworks, policy coordination, or sector-specific guidance.

This has created a global AI governance landscape that is active and fast-moving, but also fragmented. Different jurisdictions may use different definitions, risk categories, institutional models, compliance tools, and policy priorities. This fragmentation can create challenges for international cooperation, responsible innovation, public trust, and inclusive participation, especially for developing countries and smaller institutions with limited technical and regulatory capacity.

This policy brief presents a compact comparative study of selected global, regional, and national AI governance frameworks. It introduces an interpretive AI Governance Interoperability Readiness Score to compare how far different frameworks may support cooperation through shared principles, risk clarity, institutional coordination, standards linkages, international cooperation, and inclusion.

The purpose of this study is not to rank one governance model as better than another. Instead, it aims to identify patterns, areas of alignment, points of fragmentation, and practical pathways toward more inclusive and interoperable AI governance.

1. Introduction

Artificial intelligence is no longer only a technical issue. It is now a governance issue that affects rights, safety, economic development, public services, innovation, trust, and international cooperation. As AI technologies evolve rapidly, countries and regions are responding through different governance models. These include laws, strategies, ethical principles, standards, risk management frameworks, regulatory sandboxes, public-sector guidance, and multilateral initiatives.

This diversity is valuable because AI governance must reflect different legal systems, economic needs, cultural contexts, development priorities, and institutional capacities. However, when governance approaches become too disconnected, they may create confusion, duplication, uneven protections, and barriers to cooperation.

This project studies AI governance interoperability. In simple terms, interoperability means the ability of different governance systems to work together, even if they are not identical. In the context of AI governance, interoperability may include shared responsible AI principles, comparable risk categories, common standards, mutual learning, cross-border cooperation, and capacity support for countries with different levels of digital readiness.

The project examines selected AI governance frameworks and initiatives from global, regional, and national levels. It converts complex policy information into a structured mapping table, scoring model, dashboard summary, and visual outputs that can support clearer understanding by policy audiences and non-specialist readers.

2. Purpose of the Study

This study was conducted to better understand the current global AI governance landscape and explore how different approaches may support or limit interoperability.

The main objectives are to:

- 1 Map selected AI governance frameworks across global, regional, and national levels.
- 2 Compare their governance philosophy, binding status, risk approach, institutional model, standards linkages, international cooperation mechanisms, and inclusion focus.
- 3 Develop a simple interpretive scoring model to assess interoperability readiness.
- 4 Identify key patterns of alignment and fragmentation.
- 5 Present complex AI governance information in a clear and visually accessible format.
- 6 Suggest practical pathways for more inclusive and interoperable global AI governance.

The study is designed as a compact policy synthesis product. It is intended for portfolio demonstration, policy learning, and discussion support rather than legal advice or an official evaluation.

3. Frameworks Included

The study reviews selected frameworks and initiatives that represent different levels and styles of AI governance.

Level	Framework or Initiative	Main Governance Character
Global	UN High-level Advisory Body on AI	Inclusive, networked, human-centered global governance
Global	UN AI Governance for Humanity Lab	Multistakeholder cooperation and global dialogue support
Global / Intergovernmental	OECD AI Principles and OECD.AI Policy Observatory	Trustworthy AI, policy learning, and international comparability
Regional	European Union AI Act	Binding, risk-based regulation
Regional	African Union Continental AI Strategy	Development-focused, equitable, and Africa-centered AI governance
Regional	ASEAN Guide on AI Governance and Ethics	Voluntary, practical, soft-law regional coordination
National	United Kingdom Pro-Innovation AI Regulation Approach	Principles-based, regulator-led, innovation-enabling model
National / Standards	United States NIST AI Risk Management Framework	Voluntary technical risk management and assurance framework
National	China Global AI Governance Initiative and related approach	State-led governance with emphasis on security, development, and international positioning

These frameworks were selected because they reflect different governance philosophies, institutional models, geographic perspectives, and policy instruments.

Global AI Governance Patchwork Map

A visual showing global, regional, and national AI governance approaches.

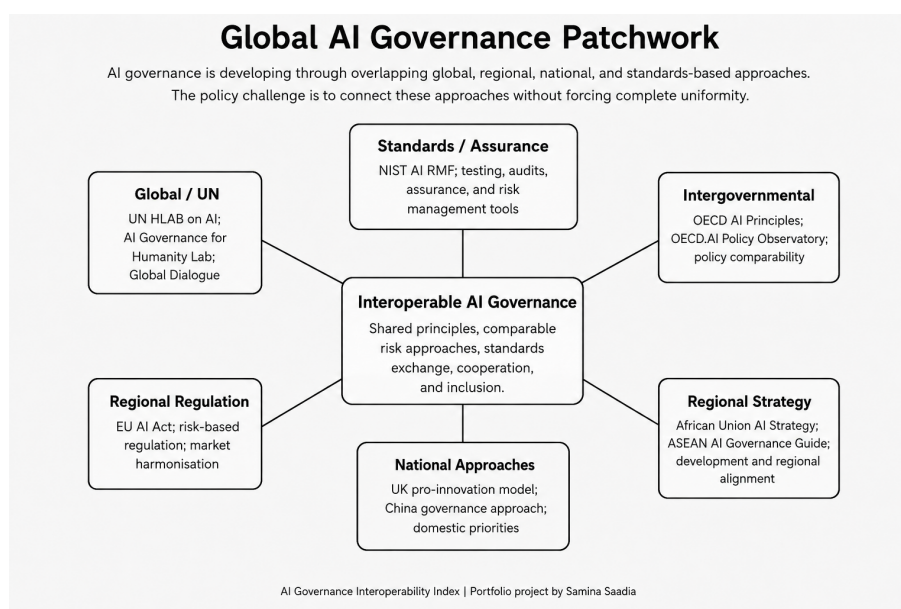


Figure 1. Global AI Governance Patchwork Map. The diagram summarizes the major governance levels and the interoperability challenge.

4. Methodology

The study uses a structured comparative mapping approach. Each selected framework is reviewed across common governance dimensions to make comparison easier.

The analysis focuses on the following questions:

- What type of governance instrument is being used?
- Is the approach binding, voluntary, strategic, or advisory?
- What is the core governance philosophy?
- How does the framework define or manage AI risk?
- Which institutions or coordination mechanisms are involved?
- Does the framework connect to standards, assurance, testing, audits, or risk management tools?
- Does it support international cooperation?
- Does it address inclusion, capacity building, and the needs of developing countries or underserved groups?
- What are its main interoperability strengths and gaps?

The scoring model is interpretive and evidence-based. It is not an official ranking. Scores are used to support structured comparison and visual analysis.

5. Scoring Dimensions

The AI Governance Interoperability Readiness Score is based on six dimensions.

Dimension	What It Measures
Shared Principles Alignment	Alignment with responsible AI principles such as safety, transparency, accountability, fairness, privacy, inclusion, and human rights
Risk Classification Clarity	Clarity of risk categories, risk assessment methods, or risk management guidance
Institutional Coordination	Presence of clear institutions, regulators, coordination bodies, or implementation mechanisms
Standards and Assurance Linkages	Connection to standards, testing, audits, impact assessments, certification, or risk management tools
International Cooperation	Support for cross-border cooperation, policy exchange, standards coordination, or multilateral dialogue
Inclusion and Capacity Building	Attention to developing countries, digital divides, marginalized groups, skills, infrastructure, and equitable participation

Each dimension is scored from 1 to 5.

Score	Meaning
1	Very limited or unclear
2	Emerging or partial
3	Moderate
4	Strong
5	Very strong

The total possible score is 30.

Interoperability Readiness Percentage = Total Score / 30 × 100

A high score does not mean a framework is perfect. A lower score does not mean a framework is unimportant. The score only indicates how strongly the framework appears to support interoperability across the selected dimensions.

5.1 Interoperability Readiness Score

The following chart summarizes the interoperability readiness percentage for each selected framework. The percentages are based on six interpretive dimensions: shared principles alignment, risk classification clarity, institutional coordination, standards and assurance linkages, international cooperation, and inclusion or capacity building.

Interoperability Readiness Score Across Selected AI Governance Frameworks

Interpretive, evidence-based policy synthesis using six interoperability dimensions

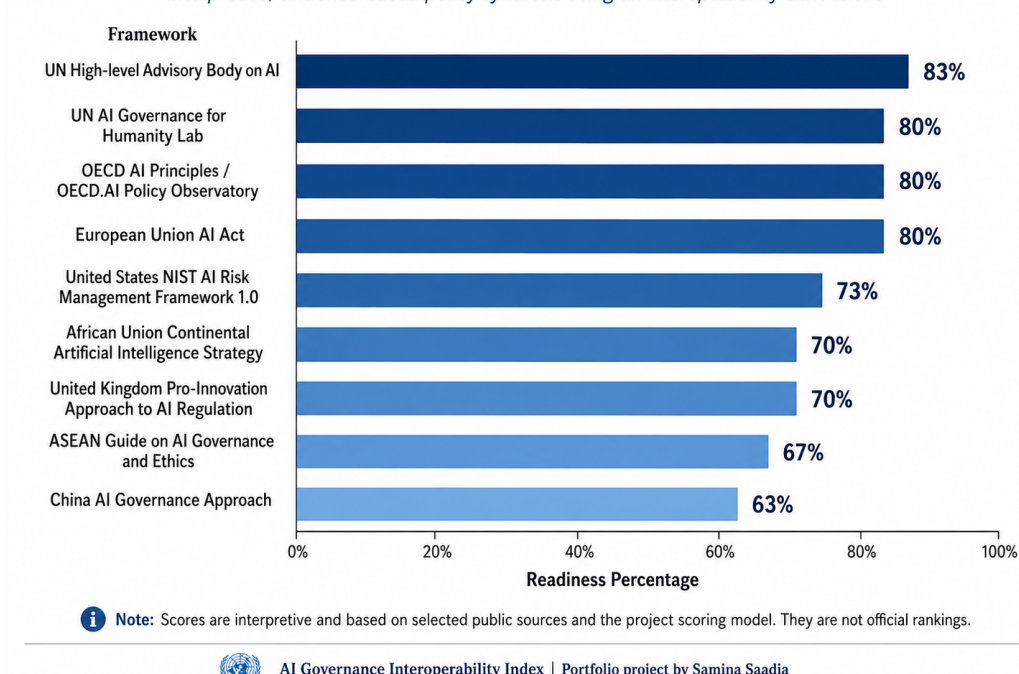


Figure 2. Interoperability Readiness Score Across Selected AI Governance Frameworks. Scores are interpretive and based on selected public sources and the project scoring model. They are intended for policy synthesis and comparison, not as official rankings.

6. Key Findings

6.1 AI governance is developing as a patchwork, not as one global model

The study shows that AI governance is emerging through different instruments and philosophies. The European Union has adopted a binding risk-based legal model. The United States NIST framework offers a voluntary risk management and assurance approach. The United Kingdom uses a principles-based and regulator-led model. ASEAN emphasizes practical and voluntary guidance. The African Union highlights development, inclusion, and regional capacity. The OECD supports international policy learning and comparability. The United Nations focuses on inclusive, networked, and human-centered global cooperation.

This diversity reflects different political, legal, economic, and institutional contexts. It is not necessarily a weakness. However, without coordination, the patchwork may become difficult for policymakers, developers, regulators, and public institutions to navigate.

Regulatory Philosophy Matrix

A matrix placing frameworks by binding versus voluntary approaches and innovation-enabling versus risk-control orientation.

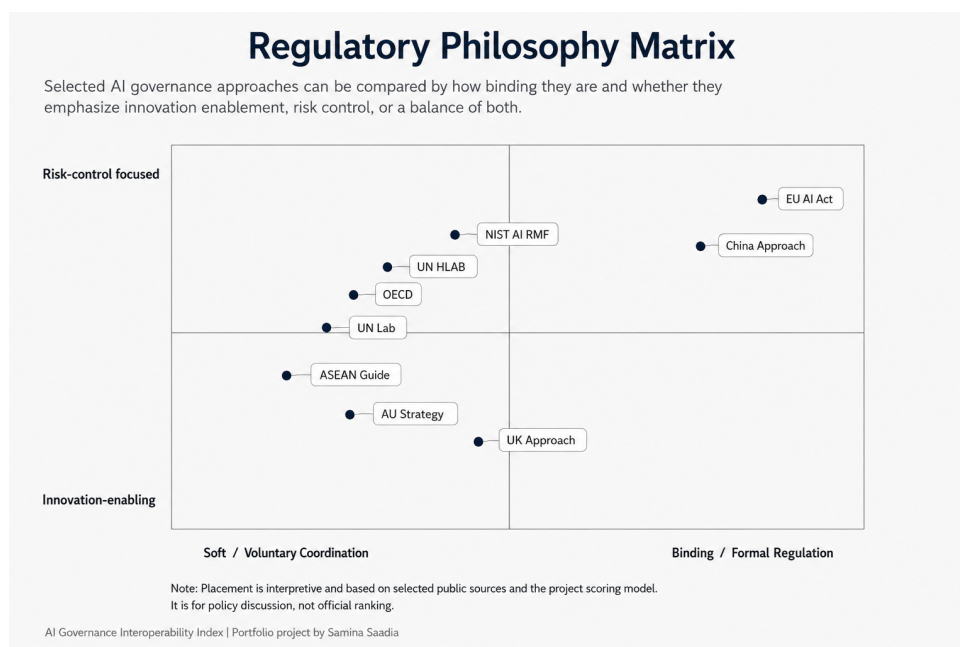


Figure 3. Regulatory Philosophy Matrix. The placement is interpretive and supports discussion rather than official ranking.

6.2 Shared principles are the strongest area of alignment

Most frameworks refer to common responsible AI principles, including safety, transparency, accountability, fairness, privacy, human oversight, and inclusion. This suggests that there is already a foundation for interoperability at the level of values and principles.

However, shared principles do not automatically create shared implementation. Different frameworks may interpret the same principle differently. For example, transparency may mean public disclosure in one context, technical documentation in another, and explainability or auditability in another. Interoperability therefore requires both common principles and clearer operational translation.

6.3 Risk governance is uneven across frameworks

Risk is central to AI governance, but frameworks vary in how clearly they classify or manage it. The EU AI Act provides a structured legal risk approach. The NIST AI Risk Management Framework provides detailed risk management guidance. Other approaches discuss risk more broadly through principles, strategies, or institutional coordination.

This unevenness creates a challenge. If risk categories, assessment methods, and responsibilities differ significantly across systems, it becomes harder to compare obligations, share lessons, or coordinate cross-border governance.

6.4 Standards and assurance tools are essential for moving from principles to practice

The study highlights the importance of standards, audits, testing, impact assessments, conformity assessments, risk management tools, and evaluation methods. These mechanisms help translate broad principles into practical implementation.

Frameworks with stronger standards and assurance linkages may be more useful for interoperability because they offer practical tools that different institutions can compare or adapt. However, standards-based approaches must also remain inclusive. Countries and institutions with limited resources may need support to participate meaningfully in standards development and implementation.

6.5 Inclusion and capacity building remain central to global legitimacy

Inclusive AI governance is not only about ethical language. It also requires meaningful participation, technical capacity, infrastructure, skills, institutional readiness, and access to governance processes. The African Union and United Nations approaches place strong emphasis on inclusion, capacity, and equitable participation.

This is important because AI governance interoperability should not only connect advanced regulatory systems. It should also support countries, communities, and institutions that have fewer resources but are still affected by AI systems and global digital markets.

6.6 International cooperation is strong in global and regional frameworks, but more limited in some national approaches

Global and regional frameworks generally give more attention to policy exchange, multilateral dialogue, and cross-border cooperation. National approaches often focus first on domestic regulatory or strategic priorities. This is understandable, but it may limit interoperability if international cooperation is not built into implementation.

The challenge is to create practical bridges between national systems and global cooperation mechanisms. These bridges may include shared terminology, policy observatories, regulatory dialogues, standards coordination, and capacity-building networks.

7. Main Interoperability Challenges

7.1 Different definitions and terminology

AI governance frameworks may use different terms for similar issues. Terms such as high-risk AI, trustworthy AI, responsible AI, safety, transparency, and accountability may vary across documents and jurisdictions. This can create confusion and make comparison difficult.

7.2 Uneven legal and institutional capacity

Some regions have advanced legal and regulatory systems, while others are still developing AI strategies, institutions, and technical expertise. This creates an implementation gap. Interoperability cannot be achieved only by aligning rules; it also requires capacity to implement and participate.

7.3 Variation between binding and voluntary approaches

Binding regulation can create clarity and enforceability, while voluntary frameworks can support flexibility and innovation. Both have value, but they may be difficult to align if expectations, obligations, and accountability mechanisms are very different.

7.4 Fragmented standards and assurance practices

Standards, audits, testing, and impact assessments are becoming increasingly important, but they are not applied uniformly. Without coordination, different assurance systems may create duplication, confusion, or exclusion.

7.5 Risk of excluding developing countries and smaller institutions

If AI governance is shaped mainly by technologically advanced economies or large institutions, global cooperation may become less inclusive. Developing countries and smaller public institutions may face difficulties in adopting complex standards or participating in global AI governance discussions.

8. Pathways Toward More Interoperable AI Governance

8.1 Develop shared policy vocabulary

A basic shared vocabulary can help policymakers and institutions compare governance approaches more easily. This does not require all countries to adopt the same laws. It requires clearer definitions of key concepts such as AI risk, transparency, accountability, human oversight, and assurance.

8.2 Strengthen risk taxonomy alignment

Frameworks should improve comparability in how they identify, classify, and manage AI risks. Shared or comparable risk taxonomies can support cooperation while still allowing jurisdictions to adapt rules to their own context.

8.3 Promote standards exchange and assurance cooperation

International cooperation should include practical exchange on standards, audits, testing methods, impact assessments, and certification approaches. This can help move responsible AI from principle-based commitments to implementable governance practice.

8.4 Build inclusive capacity support mechanisms

Interoperability should include capacity building for developing countries, public institutions, civil society, and smaller organizations. Support may include training, technical assistance, access to policy tools, participation in standards processes, and locally relevant governance guidance.

8.5 Support multistakeholder and multidisciplinary dialogue

AI governance affects many sectors and communities. Interoperability should therefore be developed through dialogue involving governments, international organizations, technical experts, civil society, academia, private sector actors, and affected communities.

8.6 Use policy observatories and comparative mapping tools

Policy observatories, mapping tools, and structured comparative datasets can help track developments and identify convergence or fragmentation. These tools can support evidence-based dialogue and reduce misunderstanding between governance systems.

8.7 Fragmentation to Interoperability Diagram

A two-sided diagram showing movement from disconnected governance systems toward shared principles, standards exchange, and cooperation.

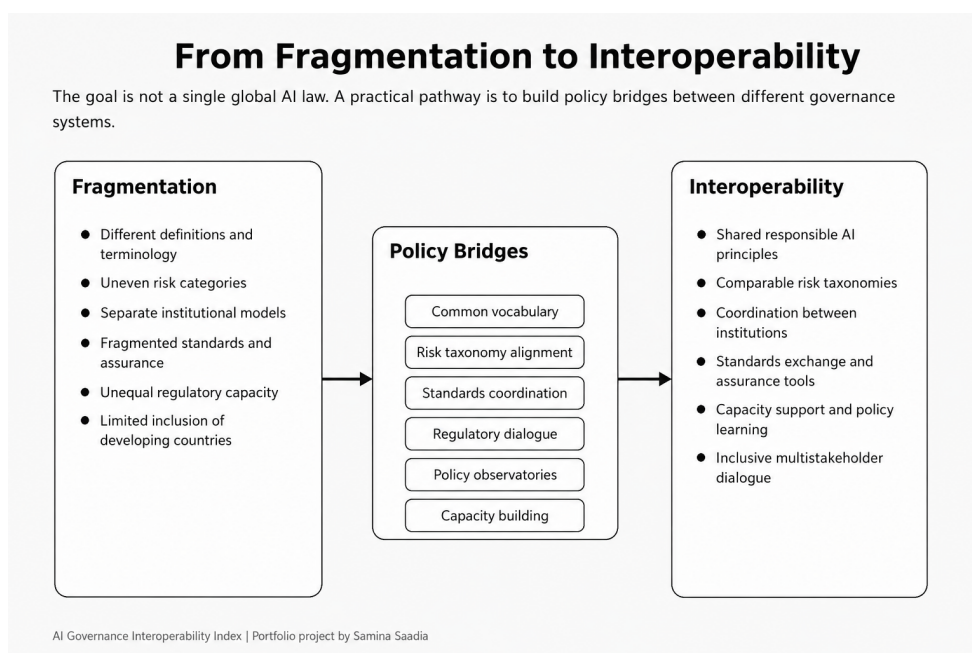


Figure 4. Fragmentation to Interoperability Diagram. The diagram summarizes practical policy bridges for improving interoperability.

9. Implications for Policy Audiences

For policymakers, the main implication is that AI governance interoperability does not require complete uniformity. Countries and regions can maintain different governance models while still cooperating through shared principles, comparable risk approaches, standards exchange, institutional dialogue, and capacity support.

For international organizations, the study highlights the importance of neutral synthesis, inclusive convening, and practical tools that make AI governance more understandable and actionable.

For developing countries and public institutions, the study shows that inclusion and capacity building must be treated as core governance issues, not secondary concerns.

For technical and policy communities, the study highlights the need to connect high-level principles with practical assurance mechanisms, including risk management, testing, monitoring, and impact assessment.

10. Conclusion

AI governance is moving quickly, but it is not moving in one single direction. Different jurisdictions and institutions are developing different models based on their legal systems, political priorities, economic needs, development goals, and risk perceptions. This diversity can support innovation and contextual relevance, but it can also create fragmentation.

The central policy challenge is not to force complete harmonization. A more realistic goal is interoperability. This means helping different governance systems communicate, compare, cooperate, and learn from one another while respecting contextual differences.

This study suggests that interoperability can be strengthened through shared principles, clearer risk classification, institutional coordination, standards and assurance linkages, international cooperation, and inclusion-focused capacity building. These pathways can help make global AI governance more coherent, practical, and inclusive.

The AI Governance Interoperability Index is a compact analytical tool for understanding this challenge. It offers a structured way to compare governance approaches, identify gaps, and support clearer policy discussion. It is not an official ranking or legal assessment, but a policy synthesis product designed to make complex AI governance information easier to understand and use.

Disclaimer

This policy brief is a compact portfolio study based on selected public sources. It does not cover every AI governance framework in the world. The scoring model is interpretive and should not be treated as an official ranking, legal opinion, or institutional assessment. The purpose is to support policy synthesis, structured comparison, and visual communication.